Adaptive Attitude Tracking of a CMG-Actuated Spacecraft with Unknown Inertia

Krittin Jindamai
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***Abstract*—This project examines an adaptive attitude tracking controller for a rigid spacecraft with three constant, unknown principal moments of inertia. An adaptive update law uses quaternion tracking errors to update inertia estimates. A pseudoinverse steering law maps the commanded torque to gimbal rates for four control moment gyroscopes (CMGs) mounted in a pyramid mounting configuration. Lyapunov stability analysis and Barbalat's lemma establish asymptotic attitude and angular velocity tracking under exact torque realization and a uniform margin from CMG singularity assumptions. Simulations are performed on MATLAB, the different gain values are tested on an oblique slew. Reducing either the feedback gain or the attitude weight increases the peak tracking errors. Reducing the adaptation gain gives the largest peaks and final inertia error. The nominal case slew gives the smallest tracking peaks, with a peak attitude error of 0.0729°. Runs that start 10° and 20° from the desired attitude converge to the reference, although their inertia estimates do not converge to the true values. When testing different motions with the same gains, periodic motion brings all three inertia estimates to the true values, while a principal-axis slew updates only the first moment of inertia.**

***Index Terms*—adaptive control, spacecraft attitude tracking, Lyapunov stability, control moment gyroscopes**

I. Introduction

A. Background and Motivation

Accurate attitude tracking is becoming more important for spacecraft proximity operations such as docking and capture. At initial contact, a docking interface can accommodate only a specified range of relative attitude and angular-rate errors [1]. This makes accuracy an important aspect that ensures these operations are safe. Formation-flying spacecraft that use laser crosslinks also require precise pointing to acquire and maintain the link between their optical terminals [2].

The inertia tensor of the spacecraft determines how the body responds to a control torque. Deploying components such as solar panels or antennas and consuming propellant can change the mass distribution and therefore the inertia [3]. Motion of liquid propellant can also introduce forces and torques that affect attitude control [4]. A controller that uses an inaccurate inertia model can therefore lose tracking accuracy. In this project, an adaptive update law uses tracking error to update the inertia estimates used in the torque command. The model treats the inertia as constant over each maneuver and unknown to the controller.

Reaction control thrusters can provide both translational and attitude control during rendezvous. During close approach, any net force from an attitude-control firing can also affect the relative trajectory and must be accounted for [5]. Reaction wheels generate torque by changing flywheel speed and are commonly used for precision pointing [5]. Control moment gyroscopes (CMGs) can deliver high torque for rapid attitude adjustments by gimbaling spinning flywheels to redirect their angular momentum [6]. This exchange of angular momentum with the spacecraft body produces torque without applying net thrust on the system. In small satellites, the actuator dynamics can also be significant because the CMG inertia can be comparable to that of the spacecraft [7].

B. Related Work and Project Scope

Ahmed et al. [3] developed a quaternion-based adaptive attitude tracking controller for a rigid spacecraft with unknown inertia. Their model takes three body-fixed torque inputs, and they used periodic reference motion to identify the six independent entries of the inertia matrix. MacKunis et al. [8] developed an adaptive attitude tracking controller for a spacecraft actuated by a four-CMG pyramid. Their model includes a time-varying inertia matrix and unknown gimbal friction, and their controller computes the gimbal rates directly using an adaptive estimate of the input matrix and a singularity-robust inverse. This project follows the approach of [3] with three unknown principal moments of inertia. The controller is the controller of [3] specialized to a diagonal inertia matrix and a scalar attitude weight, with the attitude-error gain kept as a separate design gain. Compared with [8], the CMG model is simplified. The inertia is constant, gimbal friction is neglected, and a pseudoinverse steering law maps the commanded body torque to gimbal rates. The analysis establishes asymptotic attitude and angular velocity tracking under exact torque realization and a uniform margin from CMG singularities. The simulations compare gains on a fixed slew, test convergence from initial attitude errors, and examine inertia estimation across different reference motions.

II. System Model

A. Nomenclature

The main symbols used in this paper are listed below.

F_I, F_B, F_d	Inertial, body-fixed, and desired frames
$q, q_d, q_e \in \mathbb{R}^4$	Actual, desired, and error unit quaternions (scalar last)
$\varepsilon, \varepsilon_d, \varepsilon_e$	Quaternion vector parts
η, η_d, η_e	Quaternion scalar parts
$\omega, \omega_d, \omega_e$	Actual and desired angular velocities, and angular velocity error (ω_d in F_d, the others in F_B)
R_I^B, R_I^d	Inertial-to-body and inertial-to-desired rotation matrices

C_e	Relative rotation matrix from the desired frame to the body frame
I_3	Three-dimensional identity matrix
$J \in \mathbb{R}^{3\times3}$	Spacecraft inertia matrix
$\delta, \dot{\delta} \in \mathbb{R}^4$	Gimbal angle and gimbal rate vectors
β, Ω_i	Pyramid inclination angle and flywheel speed
h, h_0	Body-frame array momentum and single-flywheel momentum magnitude
$A, A^\dagger$	Normalized CMG momentum Jacobian and its pseudoinverse
$\sigma_{\min}(A), a_{\min}$	Smallest singular value of the Jacobian and its required lower bound
$\tau_{\mathrm{cmg}}, \tau_{\mathrm{cmd}}$	Delivered and commanded body torques
r	Auxiliary tracking error
$\theta, \widehat{\theta}, \widetilde{\theta}$	Inertia parameter vector, its estimate, and estimation error
$Y(x,t)$	Inertia regressor
a, x	Known auxiliary vector, and measured attitude and angular velocity
$\bar{\omega}_d, \bar{a}_d$	Bounds on the desired angular velocity and acceleration
K, k_e	Feedback gains on the auxiliary and attitude tracking errors
α	Attitude-error weight in the auxiliary tracking error
Γ	Adaptation gain matrix
V, g	Lyapunov candidate and the nonnegative function in its derivative
z, Q	Augmented error vector and its weighting matrix
$\Delta\tau$	Torque residual between delivered and commanded torques

Throughout, $\|\cdot\|$ denotes the Euclidean norm of a vector and the induced Euclidean norm of a matrix. For a matrix, $\sigma_{\min}(\cdot)$ denotes the smallest singular value, $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ denote the smallest and largest eigenvalues, and $\succ 0$ denotes positive definiteness.

B. Kinematic Model

Using the scalar-last convention, the unit quaternion $q(t) \triangleq [\varepsilon^T(t), \eta(t)]^T \in \mathbb{R}^4$ describes the orientation of the spacecraft body-fixed frame F_B relative to the inertial frame F_I. The unit quaternion satisfies the constraint

$$\varepsilon^T\varepsilon + \eta^2 = 1. \tag{1}$$

The rotational kinematics of a rigid spacecraft body can be expressed as [3]

$$\dot{\varepsilon} = \tfrac{1}{2}\left(\eta I_3 + [\varepsilon]_\times\right)\omega, \tag{2}$$

$$\dot{\eta} = -\tfrac{1}{2}\varepsilon^T\omega. \tag{3}$$

The angular velocity $\omega(t) \in \mathbb{R}^3$, expressed in F_B, can be recovered from the quaternion kinematics as

$$\omega = 2(\eta\dot{\varepsilon} - \varepsilon\dot{\eta}) - 2[\varepsilon]_\times\dot{\varepsilon}. \tag{4}$$

Here $[\cdot]_\times$ denotes the skew-symmetric cross-product operator. For any $x = [x_1, x_2, x_3]^T \in \mathbb{R}^3$, it is defined as

$$[x]_\times = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}, \tag{5}$$

such that $[x]_\times y = x \times y$ for any $y \in \mathbb{R}^3$. It satisfies

$$[x]_\times^T = -[x]_\times, \qquad [x]_\times^2 = xx^T - \|x\|^2 I_3. \tag{6}$$

A reference frame F_d is introduced to specify the desired motion. Its orientation relative to the inertial frame is represented by the unit quaternion $q_d(t) \triangleq [\varepsilon_d^T(t), \eta_d(t)]^T \in \mathbb{R}^4$. The associated angular velocity expressed in the desired frame is $\omega_d(t) \in \mathbb{R}^3$. The rotation matrices that transform inertial coordinates to F_B and F_d, respectively, are defined as

$$R_I^B(q) = (\eta^2 - \varepsilon^T\varepsilon)I_3 + 2\varepsilon\varepsilon^T - 2\eta[\varepsilon]_\times, \tag{7}$$

$$R_I^d(q_d) = (\eta_d^2 - \varepsilon_d^T\varepsilon_d)I_3 + 2\varepsilon_d\varepsilon_d^T - 2\eta_d[\varepsilon_d]_\times. \tag{8}$$

Time arguments are omitted in the algebra when the time dependence is clear.

C. Spacecraft and CMG Dynamic Model

Assumption 1 (Rigid body dynamics and measurements): The spacecraft is modeled as a rigid body with an unknown, constant inertia matrix expressed in F_B. The selected body axes are assumed to coincide with the principal axes of the effective inertia, giving a diagonal matrix with three strictly positive unknown parameters

$$J = \begin{bmatrix} J_1 & 0 & 0 \\ 0 & J_2 & 0 \\ 0 & 0 & J_3 \end{bmatrix} \succ 0, \qquad J_1, J_2, J_3 > 0. \tag{9}$$

The attitude q and body angular velocity ω are measured.

Let $h(\delta) \in \mathbb{R}^3$ denote the CMG array's stored angular momentum, expressed in F_B. Assuming no net external torque, the dynamics of a rigid CMG-actuated spacecraft with constant inertia can be expressed as [8]

$$J\dot{\omega} = -[\omega]_\times J\omega + \tau_{\mathrm{cmg}}, \tag{10}$$

where τ_{cmg} is the net torque applied to the spacecraft body, defined as

$$\tau_{\mathrm{cmg}} = -\dot{h} - [\omega]_\times h. \tag{11}$$

D. CMG Array Model and Assumptions

The array consists of four identical single-gimbal CMGs arranged in a pyramid configuration. Each CMG maintains a constant flywheel speed $\Omega_i \in \mathbb{R}$ and a constant angular momentum magnitude $h_0 > 0$, while the gimbal angle $\delta_i(t) \in \mathbb{R}$ can change with time. The gimbal configuration is defined as

$$\delta(t) \triangleq [\delta_1(t), \delta_2(t), \delta_3(t), \delta_4(t)]^T \in \mathbb{R}^4. \tag{12}$$

With constant flywheel speed, the total CMG angular momentum derivative satisfies

$$\dot{h} = h_0 A(\delta)\dot{\delta}. \tag{13}$$

For the pyramid arrangement used in this project, shown in Fig. 1, the gimbal axes have azimuths $0, \pi/2, \pi, 3\pi/2$ and a common inclination β from the positive body z axis. At zero gimbal angle, the flywheel momentum directions are $+y, -x, -y, +x$, respectively, in body coordinates. Positive

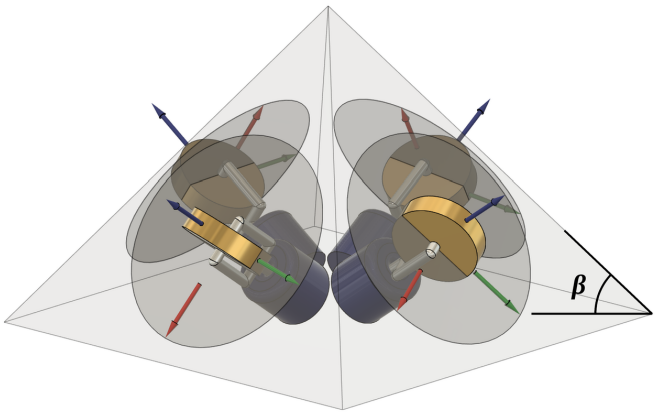


Fig. 1. Four-CMG pyramid configuration shown at nonzero gimbal angles. The angle β is the inclination of each pyramid face from the base plane (shown schematically), which equals the angle between each gimbal axis and the body z axis.

gimbal angles follow the right-hand rule about each gimbal axis. With $s_i = \sin\delta_i$, $c_i = \cos\delta_i$, $s_\beta = \sin\beta$, and $c_\beta = \cos\beta$, the total stored momentum and geometric Jacobian are [8]

$$h(\delta) = h_0 \begin{bmatrix} -c_\beta s_1 - c_2 + c_\beta s_3 + c_4 \\ c_1 - c_\beta s_2 - c_3 + c_\beta s_4 \\ s_\beta(s_1 + s_2 + s_3 + s_4) \end{bmatrix}, \tag{14}$$

$$A(\delta) = \begin{bmatrix} -c_\beta c_1 & s_2 & c_\beta c_3 & -s_4 \\ -s_1 & -c_\beta c_2 & s_3 & c_\beta c_4 \\ s_\beta c_1 & s_\beta c_2 & s_\beta c_3 & s_\beta c_4 \end{bmatrix}. \tag{15}$$

The Jacobian is obtained by differentiating $h(\delta)/h_0$ in (14) with respect to the gimbal angles. It maps gimbal rates to the normalized momentum rate in F_B. Because each CMG contributes a momentum of magnitude h_0, the total stored momentum satisfies

$$\|h(\delta)\| \le 4h_0. \tag{16}$$

This is an upper bound from the sum of four fixed-magnitude momentum vectors, rather than a guarantee against directional momentum saturation.

Assumption 2 (Actuator model): The flywheel momentum magnitude and array geometry are known, and gimbal angles are available for evaluating the actuator model. Flywheel speeds are constant, gimbal-inertia effects are negligible, and variations of effective spacecraft inertia with gimbal configuration are neglected.

Assumption 3 (Feasible torque realization): To facilitate stability analysis, the array is assumed to stay away from singular configurations by a uniform margin for all $t \ge 0$.

$$\sigma_{\min}(A(\delta(t))) \ge a_{\min} > 0. \tag{17}$$

This condition ensures that the Jacobian retains full row rank and its pseudoinverse remains bounded. The array is assumed to deliver the exact commanded torque, with gimbal-rate and travel limits inactive. Because this condition depends on the closed-loop gimbal trajectory, it cannot be verified before the motion is known. Its validity for each simulated maneuver is checked after the fact in Section VI.

III. Problem Definition and Open-Loop Error System

A. Desired Motion and Tracking Objective

Assumption 4 (Reference motion): The desired motion is described by the unit quaternion $q_d(t)$ and angular velocity $\omega_d(t)$, which satisfy the kinematics (2) and (3) with the desired quantities substituted. The desired angular velocity is assumed to be continuously differentiable, with both $\omega_d(t)$ and $\dot\omega_d(t)$ uniformly bounded for $t \ge 0$. Thus, finite constants $\bar\omega_d$ and $\bar a_d$ exist such that

$$\|\omega_d(t)\| \le \bar\omega_d, \qquad \|\dot\omega_d(t)\| \le \bar a_d. \tag{18}$$

The attitude tracking error can be defined as the rotation matrix that transforms coordinates from F_d to F_B, given by

$$C_e = R_I^B(q)\left(R_I^d(q_d)\right)^T. \tag{19}$$

Successful attitude tracking requires the actual and desired frames to align asymptotically, and their angular velocity difference to converge to zero when expressed in a common coordinate frame.

$$\lim_{t\to\infty} C_e(t) = I_3, \tag{20}$$

$$\lim_{t\to\infty} [\omega(t) - C_e(t)\omega_d(t)] = 0. \tag{21}$$

These objectives must be achieved without knowledge of the true moment of inertia. Convergence of inertia estimates to their true values is considered separately from the tracking objective.

B. Attitude and Angular Velocity Errors

The quaternion tracking error $q_e(t) \triangleq [\varepsilon_e^T(t), \eta_e(t)]^T \in \mathbb{R}^4$ is used to represent the difference between the desired and actual orientations. Its components, consistent with the convention adopted in Section II-B, are

$$\varepsilon_e = \eta_d\varepsilon - \eta\varepsilon_d + [\varepsilon]_\times\varepsilon_d, \tag{22}$$

$$\eta_e = \eta_d\eta + \varepsilon_d^T\varepsilon. \tag{23}$$

The corresponding relative rotation matrix (19) can be rewritten as

$$C_e = (\eta_e^2 - \varepsilon_e^T\varepsilon_e)I_3 + 2\varepsilon_e\varepsilon_e^T - 2\eta_e[\varepsilon_e]_\times. \tag{24}$$

Consequently, the attitude-tracking objective can be expressed as $\varepsilon_e(t) \to 0$ as $t \to \infty$. The unit norm constraint then implies $|\eta_e(t)| \to 1$. Both limiting signs represent the same physical attitude. The angular velocity tracking error is defined in the body-fixed frame as

$$\omega_e = \omega - C_e\omega_d. \tag{25}$$

Differentiating the angular velocity error gives

$$\dot\omega_e = \dot\omega + [\omega_e]_\times C_e\omega_d - C_e\dot\omega_d. \tag{26}$$

The open-loop quaternion-error kinematics are

$$\dot\varepsilon_e = \tfrac{1}{2}\left(\eta_e I_3 + [\varepsilon_e]_\times\right)\omega_e, \tag{27}$$

$$\dot\eta_e = -\tfrac{1}{2}\varepsilon_e^T\omega_e. \tag{28}$$

C. Auxiliary Tracking Error and Open-Loop Error System

To facilitate adaptive controller design and Lyapunov stability analysis, the attitude and angular velocity errors are combined into an auxiliary tracking error, defined as

$$r = \omega_e + \alpha\varepsilon_e, \tag{29}$$

where $\alpha > 0$ is a constant design gain. Differentiating (29) gives

$$\dot{r} = \dot{\omega}_e + \alpha\dot{\varepsilon}_e. \tag{30}$$

Substituting (26) into (30) and multiplying by the constant inertia matrix J yields

$$J\dot{r} = J\dot{\omega} + J[\omega_e]_\times C_e\omega_d - JC_e\dot{\omega}_d + \alpha J\dot{\varepsilon}_e. \tag{31}$$

Substituting the spacecraft dynamics (10) developed in Section II-C gives the auxiliary error dynamics

$$J\dot{r} = \tau_{\text{cmg}} - [\omega]_\times J\omega + Ja, \tag{32}$$

where a is the known auxiliary vector introduced to simplify the dynamics, expressed as

$$a = [\omega_e]_\times C_e\omega_d - C_e\dot{\omega}_d + \alpha\dot{\varepsilon}_e. \tag{33}$$

Although these dynamics are nonlinear in the measured states, the unknown diagonal inertia parameters enter linearly. Define the unknown parameter vector as $\theta = [J_1, J_2, J_3]^T$. For $\omega = [\omega_1, \omega_2, \omega_3]^T$ and $a = [a_1, a_2, a_3]^T$, the unknown inertia terms satisfy

$$-[\omega]_\times J\omega + Ja = Y(x,t)\theta, \tag{34}$$

where the regression matrix is given below and x denotes the measured attitude and angular velocity collected into one vector. The argument t represents the dependence on the prescribed reference.

$$Y(x,t) = \begin{bmatrix} a_1 & \omega_2\omega_3 & -\omega_2\omega_3 \\ -\omega_1\omega_3 & a_2 & \omega_1\omega_3 \\ \omega_1\omega_2 & -\omega_1\omega_2 & a_3 \end{bmatrix}. \tag{35}$$

Therefore, the open-loop auxiliary error system is

$$J\dot{r} = \tau_{\text{cmg}} + Y(x,t)\theta. \tag{36}$$

All quantities required to evaluate Y can be obtained from the attitude, angular velocity, prescribed reference signals, and chosen gain α. In particular, $\dot{\varepsilon}_e$ is calculated from the error kinematics (27), so measured angular acceleration is not required.

Finally, substituting the CMG torque model (11) and momentum-rate relation (13) gives the expression in terms of the gimbal-rate input

$$J\dot{r} = -h_0 A(\delta)\dot{\delta} - [\omega]_\times h(\delta) + Y(x,t)\theta. \tag{37}$$

This form separates the known actuator terms from the unknown inertia parameters and provides a starting point for the adaptive controller and steering law.

IV. Controller and Adaptive Update Law Design

A. Commanded Torque and Adaptive Update Law

Let $\widehat{\theta}(t) \in \mathbb{R}^3$ denote the estimate of the unknown inertia parameter vector. Design the commanded body torque as

$$\tau_{\text{cmd}} = -Y(x,t)\widehat{\theta} - Kr - k_e\varepsilon_e, \tag{38}$$

where $K = K^T \succ 0$ and $k_e > 0$ are constant feedback gains. The estimated-model term compensates for the inertia-dependent dynamics, while the remaining terms provide feedback through the auxiliary tracking error and the attitude tracking error. The parameter estimate is updated according to

$$\dot{\widehat{\theta}} = \Gamma Y^T(x,t)r, \qquad \widehat{\theta}(0) = \widehat{\theta}_0, \tag{39}$$

where $\Gamma = \Gamma^T \succ 0$ is a constant adaptation-gain matrix and $\widehat{\theta}_0$ is a finite initial estimate. The update law is selected to cancel the parameter-error cross term in the subsequent Lyapunov analysis. The controller, therefore, does not require inversion of the estimated inertia matrix. No filters or state observers are used.

B. CMG Steering Law

To map the requested body torque, the gimbal rates must satisfy

$$h_0 A(\delta)\dot{\delta} = -\tau_{\text{cmd}} - [\omega]_\times h(\delta). \tag{40}$$

Under the singularity margin (17) in Section II-D, $A(\delta)$ has full row rank. Consequently, its Moore–Penrose pseudoinverse is

$$A^\dagger(\delta) = A^T(\delta)\left[A(\delta)A^T(\delta)\right]^{-1}, \tag{41}$$

and satisfies

$$A(\delta)A^\dagger(\delta) = I_3. \tag{42}$$

Select the gimbal-rate command as

$$\dot{\delta}_{\text{cmd}} = -\frac{1}{h_0}A^\dagger(\delta)\left[\tau_{\text{cmd}} + [\omega]_\times h(\delta)\right]. \tag{43}$$

This is the minimum-Euclidean-norm solution of the torque-allocation equation (40). The factor of $1/h_0$ appears explicitly because the geometric Jacobian $A(\delta)$ does not include the flywheel momentum magnitude. The gimbal-rate vector is therefore the actual control input of the system, and τ_{cmd} is a commanded torque that is realized through the array.

Under the ideal gimbal-rate tracking assumption,

$$\dot{\delta} = \dot{\delta}_{\text{cmd}}. \tag{44}$$

Substituting the command (43) into the CMG torque model (11), using (13) and (42), gives

$$\begin{aligned} \tau_{\text{cmg}} &= -h_0 A(\delta)\dot{\delta}_{\text{cmd}} - [\omega]_\times h(\delta) \\ &= A(\delta)A^\dagger(\delta)\left[\tau_{\text{cmd}} + [\omega]_\times h(\delta)\right] - [\omega]_\times h(\delta) \\ &= \tau_{\text{cmd}}. \end{aligned} \tag{45}$$

Thus, the array delivers the exact commanded torque under the nonsingular and ideal-actuation assumptions established in Section II-D. MacKunis et al. [8] instead use a singularity-robust inverse, which adds a small regularization term near

singular configurations. When their control input is substituted into the error system, the product of the input matrix and its regularized inverse is also treated as the identity [8].

C. Closed-Loop Error System

The parameter estimation error is defined as

$$\widetilde{\theta}(t) = \theta - \widehat{\theta}(t). \tag{46}$$

Because the true-inertia parameters are constant, the adaptive update law gives

$$\dot{\widetilde{\theta}} = -\dot{\widehat{\theta}}. \tag{47}$$

Substituting the torque realization result (45) from Section IV-B into the open-loop auxiliary error system (36) and applying the torque controller (38) yields

$$J\dot{r} = \tau_{\mathrm{cmd}} + Y(x,t)\theta, \tag{48}$$

$$J\dot{r} = -Y(x,t)\widehat{\theta} - Kr - k_e\varepsilon_e + Y(x,t)\theta. \tag{49}$$

The resulting closed-loop equations are therefore

$$J\dot{r} = Y(x,t)\widetilde{\theta} - Kr - k_e\varepsilon_e, \tag{50}$$

$$\dot{\widetilde{\theta}} = -\Gamma Y^T(x,t)r. \tag{51}$$

Together with the quaternion-error kinematics in Section III-B, these equations define the closed-loop system used in the stability analysis. Since $\omega = r - \alpha\varepsilon_e + C_e\omega_d$ and $\dot{\varepsilon}_e$ depends only on q_e and ω_e, the regressor can be written as a function of q_e, r, and t. The error states $(q_e, r, \widetilde{\theta})$ therefore form a closed system, and the gimbal angles follow from the steering law without affecting the tracking errors.

V. Stability Analysis

A. Supporting Results and Main Theorem

For a function $f : [0,\infty) \to \mathbb{R}^n$, the notation $f \in \mathcal{L}_\infty$ indicates boundedness, while $f \in \mathcal{L}_2$ means the function is square integrable.

$$\int_0^\infty \|f(t)\|^2\, dt \in \mathcal{L}_\infty. \tag{52}$$

Lemma 1 (Lyapunov-like form of Barbalat's lemma): Let $V : [0,\infty) \to \mathbb{R}$ be continuously differentiable and bounded below. If $\dot{V}(t) \le -g(t)$, where g is nonnegative and uniformly continuous, then $g(t) \to 0$ as $t \to \infty$. Integrating the derivative inequality bounds the integral of g, so the result follows from [9, Lemma 8.2]. The equality case $g = -\dot{V}$ is the Lyapunov-like form in [10, Lemma 4.3].

Lemma 2 (Rayleigh–Ritz bounds): Let $Q \in \mathbb{R}^{n\times n}$ be a symmetric positive definite matrix. Let $x \in \mathbb{R}^n$ and $V(x) = x^TQx$. If $\lambda_{\min}(Q)$ and $\lambda_{\max}(Q)$ are the minimum and maximum eigenvalues of Q, respectively, then $\lambda_{\min}(Q)\|x\|^2 \le V(x) \le \lambda_{\max}(Q)\|x\|^2$ for every $x \in \mathbb{R}^n$ [11, Corollary 9.3].

Theorem 1 (Asymptotic tracking under feasible CMG operation): Consider the closed-loop system (50)–(51). Suppose Assumptions 1–4 hold and the gains satisfy

$$K = K^T \succ 0, \quad \Gamma = \Gamma^T \succ 0, \quad k_e > 0, \quad \alpha > 0. \tag{53}$$

In particular, the actuator assumptions require exact torque realization and the uniform singularity margin (17) along the considered motion. For finite initial states and parameter estimates, with unit initial quaternions, all closed-loop signals remain bounded, and

$$r(t) \to 0, \quad \varepsilon_e(t) \to 0, \quad \omega_e(t) \to 0, \quad C_e(t) \to I_3. \tag{54}$$

These conclusions are conditional on the actuator assumptions. Convergence of the inertia estimates to their true values is not asserted.

B. Lyapunov Candidate and Its Properties

Consider the candidate Lyapunov function

$$V = k_e\left[(1-\eta_e)^2 + \varepsilon_e^T\varepsilon_e\right] + \tfrac{1}{2}r^TJr + \tfrac{1}{2}\widetilde{\theta}^T\Gamma^{-1}\widetilde{\theta}. \tag{55}$$

Defining the augmented error vector as $z = [\varepsilon_e^T, 1-\eta_e, r^T, \widetilde{\theta}^T]^T$, the Lyapunov candidate can be written as

$$V(z) = z^TQz, \tag{56}$$

where

$$Q = \begin{bmatrix} k_eI_3 & 0 & 0 & 0 \\ 0 & k_e & 0 & 0 \\ 0 & 0 & \frac{1}{2}J & 0 \\ 0 & 0 & 0 & \frac{1}{2}\Gamma^{-1} \end{bmatrix} \in \mathbb{R}^{10\times 10}. \tag{57}$$

Since $J \succ 0$, $\Gamma^{-1} \succ 0$, and $k_e > 0$, Q is symmetric positive definite. By the Rayleigh–Ritz theorem (Lemma 2), the candidate satisfies

$$c_1\|z\|^2 \le V(z) \le c_2\|z\|^2, \qquad z \in \mathbb{R}^{10}, \tag{58}$$

where $c_1 = \lambda_{\min}(Q) > 0$ and $c_2 = \lambda_{\max}(Q) > 0$. Thus, V is positive definite relative to the augmented error equilibrium $(\varepsilon_e, \eta_e, r, \widetilde{\theta}) = (0,1,0,0)$ and is lower bounded by zero. Furthermore, the lower bound in (58) implies $V \to \infty$ as $\|z\| \to \infty$ along admissible error states. Because the quaternion components satisfy the unit-norm constraint, this radial-growth property concerns the unbounded components r and $\widetilde{\theta}$. The gimbal angles are treated separately. The constants c_1 and c_2 are required only for the analysis. Their numerical values, including their dependence on the unknown true inertia, need not be available to the controller.

C. Lyapunov Derivative and Cross Term Cancellation

The unit quaternion constraint gives

$$(1-\eta_e)^2 + \varepsilon_e^T\varepsilon_e = 1 - 2\eta_e + \eta_e^2 + \varepsilon_e^T\varepsilon_e = 2(1-\eta_e). \tag{59}$$

Using the quaternion-error kinematics (28) and $\omega_e = r - \alpha\varepsilon_e$, the derivative of the attitude term is

$$\dot{V}_{\mathrm{att}} = -2k_e\dot{\eta}_e = k_e\varepsilon_e^T\omega_e = k_e\varepsilon_e^Tr - k_e\alpha\|\varepsilon_e\|^2. \tag{60}$$

Since J is constant and symmetric, substitution of the closed-loop error dynamics (50) gives

$$\dot{V}_r = r^TJ\dot{r} = r^TY\widetilde{\theta} - r^TKr - k_er^T\varepsilon_e. \tag{61}$$

Similarly, since Γ is constant and symmetric, the parameter-error term satisfies

$$\dot{V}_\theta = \widetilde{\theta}^T \Gamma^{-1} \dot{\widetilde{\theta}} = -\widetilde{\theta}^T Y^T r = -r^T Y \widetilde{\theta}. \tag{62}$$

Combining the three contributions gives the following expression. The parameter-error terms cancel and the attitude cross terms also cancel.

$$\begin{aligned} \dot{V} &= k_e \varepsilon_e^T r - k_e \alpha \|\varepsilon_e\|^2 + r^T Y \widetilde{\theta} \\ &\quad - r^T K r - k_e r^T \varepsilon_e - r^T Y \widetilde{\theta} \\ &= -k_e \alpha \|\varepsilon_e\|^2 - r^T K r \le 0. \end{aligned} \tag{63}$$

The derivative in (63) is negative semidefinite in the augmented error z. It can vanish while $\widetilde{\theta} \neq 0$, so this derivative bound alone does not establish convergence of the complete augmented error. Define the nonnegative function

$$g(t) = r^T K r + k_e \alpha \|\varepsilon_e\|^2 \ge 0. \tag{64}$$

Then $\dot{V} = -g(t)$, and $0 \le V(t) \le V(0) = V_0$ for all $t \ge 0$.

D. Boundedness of Closed-Loop Signals

The bounds proceed from the tracking and parameter errors to the regressor, torque, and gimbal rates.

Tracking and parameter error bounds. From the Lyapunov candidate (55) and $V(t) \le V_0$,

$$\|r(t)\| \le \sqrt{\frac{2V_0}{\lambda_{\min}(J)}}, \qquad \|\widetilde{\theta}(t)\| \le \sqrt{\frac{2V_0}{\lambda_{\min}(\Gamma^{-1})}}. \tag{65}$$

The unit quaternion constraint gives

$$\|\varepsilon_e(t)\| \le 1, \qquad |\eta_e(t)| \le 1. \tag{66}$$

Furthermore, since $\widehat{\theta} = \theta - \widetilde{\theta}$, we have $\|\widehat{\theta}(t)\| \le \|\theta\| + \|\widetilde{\theta}(t)\|$. Since θ is finite and constant, the parameter estimates are bounded.

Angular velocity and quaternion derivative bounds. The definitions of the tracking errors imply

$$\|\omega_e(t)\| \le \|r(t)\| + \alpha \|\varepsilon_e(t)\|, \qquad \|\omega(t)\| \le \|\omega_e(t)\| + \bar{\omega}_d. \tag{67}$$

Both ω_e and ω are therefore bounded. The second inequality uses the fact that the relative rotation preserves the Euclidean norm of the desired angular velocity. From the quaternion-error kinematics (27)–(28),

$$\|\dot{\varepsilon}_e(t)\| \le \tfrac{1}{2} \left(|\eta_e(t)| + \|\varepsilon_e(t)\| \right) \|\omega_e(t)\| \le \|\omega_e(t)\|, \tag{68}$$

$$|\dot{\eta}_e(t)| \le \tfrac{1}{2} \|\varepsilon_e(t)\| \, \|\omega_e(t)\|. \tag{69}$$

Regressor bound. Both quaternion-error derivatives are therefore bounded. Recall from Section III that

$$a = [\omega_e]_\times C_e \omega_d - C_e \dot{\omega}_d + \alpha \dot{\varepsilon}_e. \tag{70}$$

The reference bounds (18) imply

$$\|a(t)\| \le \|\omega_e(t)\| \bar{\omega}_d + \bar{a}_d + \alpha \|\dot{\varepsilon}_e(t)\|. \tag{71}$$

Thus, $a(t)$ is bounded. For diagonal inertia, the regressor satisfies

$$Y(x,t) = -[\omega]_\times \mathrm{diag}(\omega) + \mathrm{diag}(a). \tag{72}$$

Using $\|[\omega]_\times\| = \|\omega\|$ and $\|\mathrm{diag}(v)\| \le \|v\|$ for the induced Euclidean matrix norm gives

$$\|Y(x,t)\| \le \|\omega\|^2 + \|a\|. \tag{73}$$

Therefore, $Y(x,t)$ is bounded.

Torque and error derivative bounds. The commanded torque (38) satisfies

$$\|\tau_{\mathrm{cmd}}\| \le \|Y\| \, \|\widehat{\theta}\| + \|K\| \, \|r\| + k_e \|\varepsilon_e\|. \tag{74}$$

From the closed-loop auxiliary error dynamics (50),

$$\|\dot{r}\| \le \|J^{-1}\| \left(\|Y\| \, \|\widetilde{\theta}\| + \|K\| \, \|r\| + k_e \|\varepsilon_e\| \right). \tag{75}$$

The adaptive law (39) also gives

$$\|\dot{\widehat{\theta}}\| \le \|\Gamma\| \, \|Y\| \, \|r\|. \tag{76}$$

Since every term on the right is bounded, τ_{cmd}, $\dot{r}$, and $\dot{\widehat{\theta}}$ are also bounded. Exact torque realization gives bounded delivered torque. These derivative bounds are analytical properties used in the proof and do not require measurement of $\dot{r}$ or actual angular acceleration.

Gimbal rate bound. The Euclidean induced norm of the pseudoinverse is the reciprocal of the smallest singular value of the full-row-rank Jacobian. The uniform singularity margin (17) gives

$$\|A^\dagger(\delta(t))\| = \frac{1}{\sigma_{\min}(A(\delta(t)))} \le \frac{1}{a_{\min}}. \tag{77}$$

Using the steering law (43) and the momentum bound (16),

$$\begin{aligned} \|\dot{\delta}_{\mathrm{cmd}}\| &\le \frac{1}{h_0} \|A^\dagger(\delta)\| \left(\|\tau_{\mathrm{cmd}}\| + \|\omega\| \, \|h\| \right) \\ &\le \frac{\|\tau_{\mathrm{cmd}}\| + 4h_0 \|\omega\|}{h_0 a_{\min}}. \end{aligned} \tag{78}$$

Thus, the commanded and actual gimbal rates are bounded under ideal rate tracking.

Signal chasing. Combining the bounds above, standard signal chasing arguments show that all closed-loop signals remain bounded. In particular,

$$r, \varepsilon_e, \widetilde{\theta}, \widehat{\theta}, \omega_e, \omega, a, Y, \tau_{\mathrm{cmd}}, \dot{r}, \dot{\varepsilon}_e, \dot{\widehat{\theta}}, \dot{\delta} \in \mathcal{L}_\infty. \tag{79}$$

Implementability. The measured attitude q and angular velocity ω, together with the supplied reference signals q_d, ω_d, and $\dot{\omega}_d$, determine the tracking errors and the regressor $Y(x,t)$. The quaternion error derivative is computed from (27), and the estimate $\widehat{\theta}$ is generated by integrating (39) from its specified initial value. The controller (38) and adaptive update law (39) are therefore implementable with the available signals, the estimate, and the selected gains. The commanded torque and adaptation rate are bounded by (74) and (76), respectively.

The measured gimbal angles and the known array model give $A(\delta)$ and $h(\delta)$. Under the singularity margin in Assumption 3, the steering law (43) gives a computable gimbal-rate command whose bound is given in (78).

E. Integrability, Uniform Continuity, and Convergence

Lower boundedness and integrability. The candidate satisfies $V(t) \geq 0$, and its derivative satisfies $\dot{V}(t) = -g(t) \leq 0$. Integrating gives

$$\int_0^T g(t)\,dt = V(0) - V(T) \leq V_0. \tag{80}$$

Because $g(t) \geq 0$, this integral is nondecreasing in T and bounded above by V_0. Taking $T \to \infty$ gives

$$\int_0^\infty g(t)\,dt \leq V_0 < \infty. \tag{81}$$

Moreover, by Rayleigh–Ritz (Lemma 2),

$$r^T K r \geq \lambda_{\min}(K)\|r\|^2.$$

Both terms in (64) are nonnegative, so integrating each term and using (81) gives

$$\int_0^\infty \|r(t)\|^2\,dt \leq \frac{V_0}{\lambda_{\min}(K)}, \quad \int_0^\infty \|\varepsilon_e(t)\|^2\,dt \leq \frac{V_0}{k_e\alpha}. \tag{82}$$

Combining these estimates with Section V-D establishes

$$r, \varepsilon_e \in \mathcal{L}_2 \cap \mathcal{L}_\infty. \tag{83}$$

Uniform continuity. Differentiating (64) and using the symmetry of the constant matrix K yields

$$\dot{g}(t) = 2r^T(t)K\dot{r}(t) + 2k_e\alpha\varepsilon_e^T(t)\dot{\varepsilon}_e(t). \tag{84}$$

Therefore,

$$|\dot{g}| \leq 2\|K\|\,\|r\|\,\|\dot{r}\| + 2k_e\alpha\|\varepsilon_e\|\,\|\dot{\varepsilon}_e\|. \tag{85}$$

Each term on the right is bounded by Section V-D. Therefore, $\dot{g}$ is bounded and g is uniformly continuous. Since $\dot{V} = -g$, the Lyapunov derivative is uniformly continuous as well.

Application of Barbalat's lemma. Since V is lower bounded, $\dot{V} = -g \leq 0$, and g is uniformly continuous, Lemma 1 gives

$$g(t) \to 0, \qquad \dot{V}(t) \to 0 \qquad \text{as} \qquad t \to \infty. \tag{86}$$

Using Rayleigh–Ritz again,

$$g(t) \geq \lambda_{\min}(K)\|r(t)\|^2 + k_e\alpha\|\varepsilon_e(t)\|^2. \tag{87}$$

Both terms on the right are nonnegative, with strictly positive coefficients. Since $g(t) \to 0$,

$$r(t) \to 0, \qquad \varepsilon_e(t) \to 0. \tag{88}$$

The auxiliary tracking error definition (29) then gives

$$\|\omega_e(t)\| \leq \|r(t)\| + \alpha\|\varepsilon_e(t)\| \to 0. \tag{89}$$

The unit-quaternion constraint gives $\eta_e^2(t) = 1 - \|\varepsilon_e(t)\|^2 \to 1$. Subtracting the identity matrix from the relative rotation matrix (24) yields

$$C_e - I_3 = -2\|\varepsilon_e\|^2 I_3 + 2\varepsilon_e\varepsilon_e^T - 2\eta_e[\varepsilon_e]_\times. \tag{90}$$

Using the triangle inequality and the norm of the cross-product matrix gives

$$\|C_e - I_3\| \leq 4\|\varepsilon_e\|^2 + 2|\eta_e|\|\varepsilon_e\| \to 0. \tag{91}$$

Therefore, $C_e(t) \to I_3$ and $\omega_e(t) \to 0$. These limits satisfy the tracking objectives (20) and (21), completing the proof of Theorem 1.

F. Corollary and Interpretation

Corollary 1 (Additional closed-loop properties): Under the conditions of Theorem 1, the selected augmented error equilibrium $z = 0$ is uniformly Lyapunov stable within the class of motions satisfying the actuator assumptions. The auxiliary tracking error and quaternion vector error belong to $\mathcal{L}_2 \cap \mathcal{L}_\infty$. The parameter estimates, commanded and delivered torques, and gimbal rates remain bounded. The adaptation rate tends to zero.

Proof: For any initial time t_0, (58) and $\dot{V} \leq 0$ give

$$c_1\|z(t)\|^2 \leq V(t) \leq V(t_0) \leq c_2\|z(t_0)\|^2. \tag{92}$$

Taking square roots gives

$$\|z(t)\| \leq \sqrt{\frac{c_2}{c_1}}\,\|z(t_0)\|, \qquad t \geq t_0. \tag{93}$$

For any $\epsilon > 0$, choosing $\|z(t_0)\| < \epsilon\sqrt{c_1/c_2}$ ensures $\|z(t)\| < \epsilon$ for all $t \geq t_0$. This choice is independent of t_0, establishing uniform Lyapunov stability. The boundedness and square-integrability claims follow from Sections V-D and V-E, including (82). Finally, by (76),

$$\|\dot{\hat{\theta}}(t)\| \leq \|\Gamma\|\,\|Y(x,t)\|\,\|r(t)\| \to 0. \tag{94}$$

This follows because Y is bounded and $r(t) \to 0$ by (88). ■

Tracking rate. Theorem 1 establishes asymptotic tracking. Establishing an exponential decay rate would require a separate argument.

Parameter estimation. The analysis establishes bounded estimation error and a vanishing adaptation rate. Convergence of the estimates to a constant or to the true inertia requires further analysis of the excitation and identifiability of the parameters.

Quaternion representation. The limit $|\eta_e(t)| \to 1$ allows either quaternion sign, both of which represent the same physical attitude. The uniform stability bound concerns the selected representation $\eta_e = 1$.

CMG feasibility. Theorem 1 assumes exact torque realization and a uniform singularity margin. For a simulated maneuver, these conditions can be assessed from the torque error and the smallest singular value of the Jacobian over the maneuver. If the delivered torque differs from the commanded torque by $\Delta\tau = \tau_{\text{cmg}} - \tau_{\text{cmd}}$, the Lyapunov derivative becomes

$$\dot{V} = -g(t) + r^T\Delta\tau. \tag{95}$$

Convergence in the presence of torque error requires a separate analysis of the additional term.

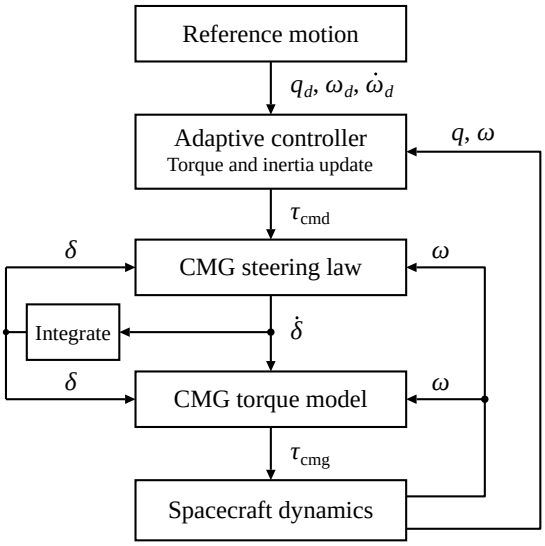


Fig. 2. Closed-loop spacecraft simulation with the four-CMG array.

VI. Numerical Simulations

A. Simulation Setup and Test Cases

The simulations use the rigid spacecraft and the CMG pyramid configuration described in Section II. The block diagram, Fig. 2, shows how the adaptive controller uses the desired motion, the current spacecraft attitude, and angular velocity to compute the commanded torque and update the inertia estimates. The steering law uses the gimbal angles and body angular velocity to compute the gimbal rates. The torque produced by the CMG array is applied to the spacecraft dynamics.

Table I lists the representative spacecraft and CMG parameters, initial conditions, and integration settings. The spacecraft starts from rest with all four gimbal angles set to zero. Except in the initial-error cases, it also starts at the desired initial attitude. The plant uses the true moments of inertia, and the controller starts with the same estimate for each principal moment. MATLAB's ode45 integrates the spacecraft states, gimbal angles, inertia estimates, and desired quaternion together for 100 s, with relative and absolute tolerances of 10^{-10} and 10^{-12}. The desired angular velocity is specified in the desired frame and integrated through the quaternion kinematics to obtain q_d. The tolerances are tighter than the ode45 defaults because the tracking errors during the hold become smaller than the default absolute tolerance.

Table II gives the three reference motions. The oblique slew turns the spacecraft 25° about an axis with equal components along all three body axes and serves as the baseline for the gain comparison. The principal-axis slew makes the same turn about the first body axis to examine which inertia estimates are updated. Both slews takes 30 s to complete with zero desired rate and acceleration at the endpoints, followed by a 70 s hold to observe how the tracking errors settle.

TABLE I
Spacecraft, actuator, initial conditions, and integration settings

Quantity	Value
True inertia J	$\mathrm{diag}(0.38, 0.37, 0.22)$ kg m^2
CMG geometry and spin	Four CMGs in a pyramid, $\beta = 54.74^\circ$, $\Omega_i = 838$ rad/s
Flywheel inertia and momentum	$I_w = 4.2 \times 10^{-5}$ kg m^2, $h_0 = 0.035196$ N m s
Singularity threshold	$a_{\min} = 0.20$ for the geometric Jacobian $A(\delta)$
Actual and desired quaternions	$q_d(0) = [0, 0, 0, 1]^T$, and $q(0) = q_d(0)$ except in the initial-error cases
Initial attitude offsets	10° and 20° about $[1, -2, 1]^T/\sqrt{6}$
Initial body rate and gimbal angles	$\omega(0) = 0$ and $\delta(0) = 0$
Initial inertia estimate	$\widehat{\theta}(0) = [0.26, 0.26, 0.26]^T$ kg m^2
Integration settings	MATLAB ode45, RelTol 10^{-10}, AbsTol 10^{-12}, 100 s run, outputs every 0.05 s

TABLE II
Commanded reference motions

Reference	Definition
Oblique slew	25° about $[1, 1, 1]^T/\sqrt{3}$ in 30 s, then hold to 100 s
Principal-axis slew	25° about $[1, 0, 0]^T$ in 30 s, then hold to 100 s
Periodic motion	$\omega_d(t) = 0.035\,[\sin(0.4t), \sin(0.8t), \sin(1.2t)]^T$ rad/s throughout the run

The periodic reference commands sinusoidal rates at three different frequencies. It keeps all three axes in motion for 100 s to test inertia estimation under sustained excitation.

The initial-error cases use the oblique slew with the spacecraft rotated away from the desired attitude at $t = 0$. The initial attitude is offset by 10° or 20° about the fixed axis $[1, -2, 1]^T/\sqrt{6}$, and the spacecraft still starts from rest. The other runs start with zero tracking error, so their errors come only from the inertia mismatch. These cases test convergence from a nonzero initial tracking error.

Table III gives the gains. The slew comparison reduces K, Γ, or α one at a time from the nominal case. The initial-error and motion cases use the nominal gains. The adaptation gain is reduced by a larger factor than K and α because the regressor is small for these slow maneuvers (Section VII-A).

B. Nominal Slew and Control Input

The attitude tracking error is reported as the physical rotation angle $2\,\mathrm{atan2}(\|\varepsilon_e\|, |\eta_e|)$, and the angular velocity tracking error is reported as $\|\omega_e\|$. Fig. 3 shows the nominal oblique slew. The commanded torque first accelerates the spacecraft and then changes direction to brake the rotation. The attitude error is largest during the slew and becomes small during the hold. The gimbal rates are the actual control input and follow the same accelerate-then-brake pattern.

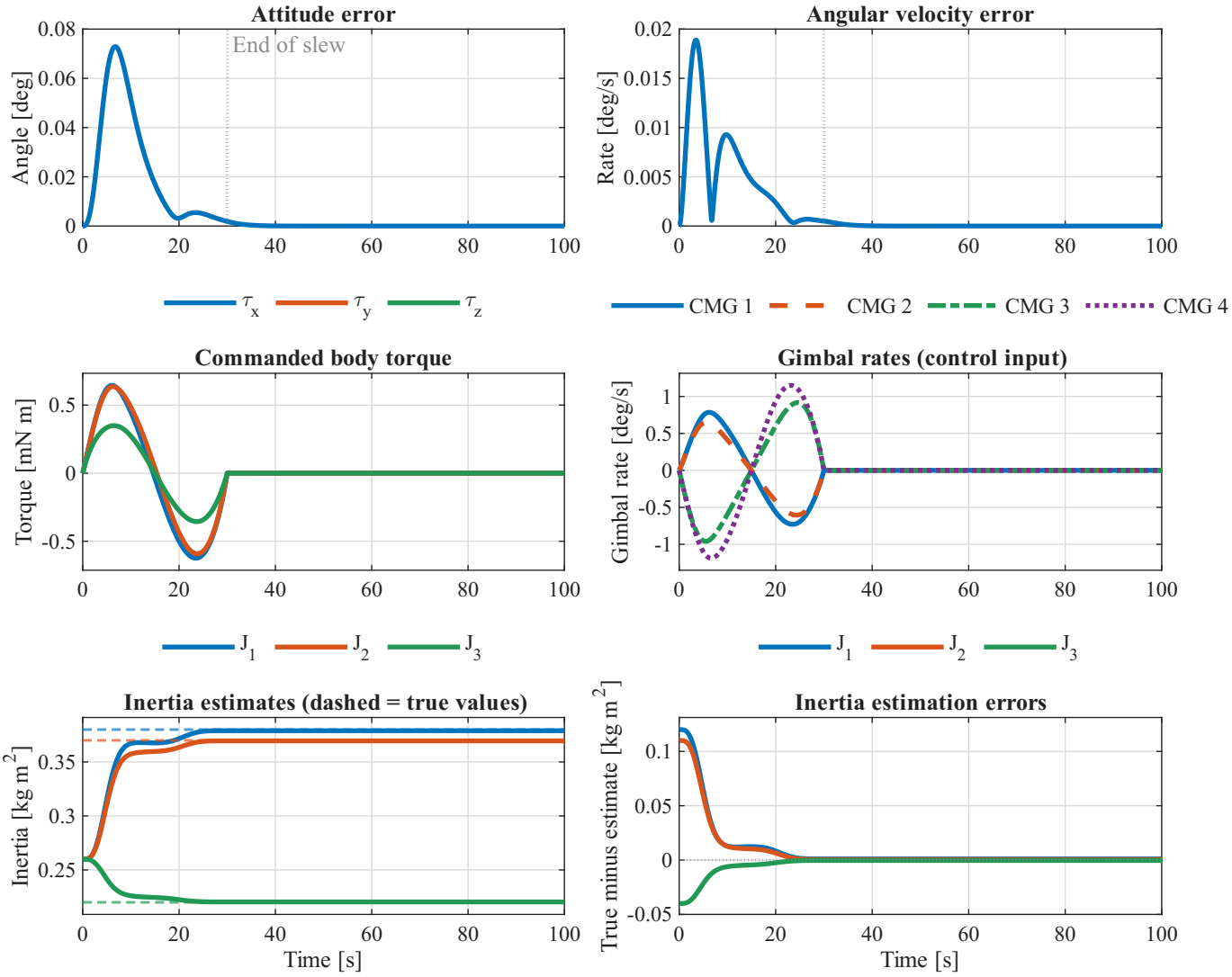


Fig. 3. Nominal oblique slew. Dashed lines mark the true moments of inertia.

TABLE III
CONTROL AND ADAPTATION GAINS BY CASE

Case	Reference	K	Γ	α	k_e
Nominal	Oblique slew	$0.30I_3$	$40{,}000I_3$	0.45	0.05
Lower K	Oblique slew	$0.18I_3$	$40{,}000I_3$	0.45	0.05
Lower Γ	Oblique slew	$0.30I_3$	$350I_3$	0.45	0.05
Lower α	Oblique slew	$0.30I_3$	$40{,}000I_3$	0.30	0.05
10°, 20° offsets	Oblique slew	$0.30I_3$	$40{,}000I_3$	0.45	0.05
Principal-axis	Principal-axis	$0.30I_3$	$40{,}000I_3$	0.45	0.05
Periodic motion	Periodic	$0.30I_3$	$40{,}000I_3$	0.45	0.05

TABLE IV
PEAK TRACKING ERRORS, SETTLING TIME, AND FINAL INERTIA ESTIMATION ERROR. THE SETTLING TIME IS THE LAST TIME THE ATTITUDE ERROR EXCEEDS 0.001°.

Case	Peak att. error [deg]	Peak rate error [deg/s]	Settling time [s]	Final inertia error [kg m^2]
Nominal	0.0729	0.0189	32.10	1.22×10^{-3}
Lower K	0.0926	0.0253	28.85	4.55×10^{-4}
Lower Γ	0.1446	0.0259	41.95	0.161
Lower α	0.0863	0.0210	34.65	1.53×10^{-3}
10° offset	10.00	2.480	49.00	0.165
20° offset	20.00	5.778	46.20	0.0997
Principal-axis slew	0.0536	0.0189	18.55	0.117
Periodic motion	0.0208	0.0604	13.80	$< 10^{-10}$

As shown in Table IV, the nominal peak attitude and angular velocity errors are 0.0729° and 0.0189°/s. The attitude error stays below 0.001° after 32.1 s, and both errors continue to decrease during the hold. The three inertia estimates approach their true values during the maneuver and change little during the hold. The final inertia estimation error norm is 1.22×10^{-3} kg m^2.

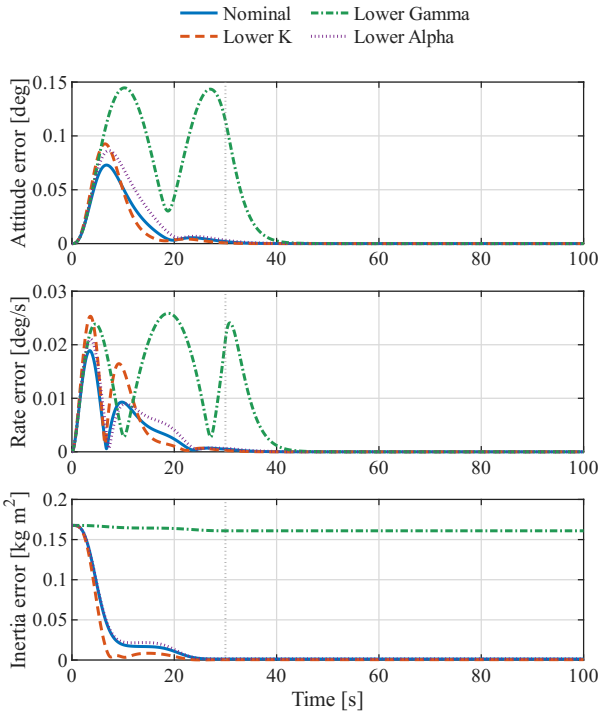


Fig. 4. Oblique slew gain comparison. The dotted vertical line marks the start of the attitude hold.

C. Slew Gain Comparison

The four runs in Fig. 4 use the same oblique reference and initial conditions. Each lower-gain case changes one setting from the nominal case, so the differences in the tracking and estimation curves can be associated with that change.

The nominal setting gives the smallest peak attitude and angular velocity errors of the four runs (Table IV). Reducing Γ gives the largest peak errors, the longest settling time of 41.95 s, and a final inertia estimation error of 0.161 kg m^2. The lower-K case has the smallest final inertia error, 4.55×10^{-4} kg m^2.

D. Initial Attitude Error

Fig. 5 shows the 10° and 20° initial-error cases on a logarithmic scale. In both runs, the attitude error decreases from its initial value, rises slightly when the slew decelerates near 25 s, and then decreases at a nearly constant rate during the hold. The attitude error stays below 0.001° after 49.0 s in the 10° case and after 46.2 s in the 20° case (Table IV). The Lyapunov function V in Fig. 6 decreases throughout both runs, consistent with (63). It levels off at a positive value because the parameter estimation error does not converge to zero.

The inertia estimates do not converge to the true values in these runs. In the 10° case, the estimate of J_2 rises to 1.85 kg m^2 within 3 s, and each of the three estimates becomes negative at some point between 0.3 s and 1.7 s. The estimates then settle at 0.407, 0.533, and 0.226 kg m^2 (Fig. 6 and Table V). The final inertia error norm is 0.165 kg m^2 for the 10° case and 0.0997 kg m^2 for the 20° case.

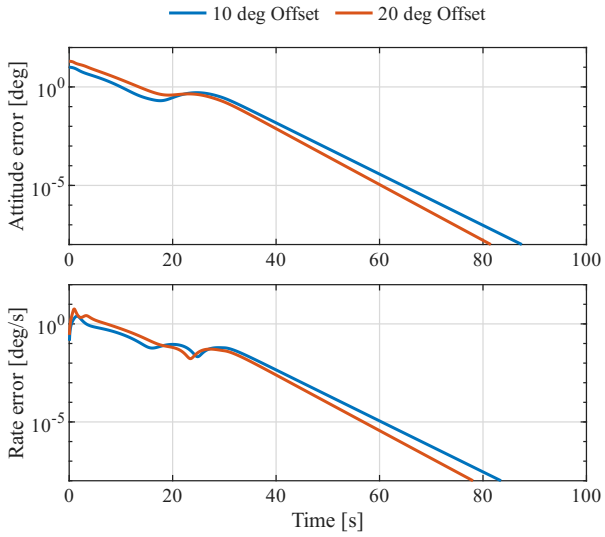


Fig. 5. Attitude and angular velocity errors for the 10° and 20° initial-error cases at the nominal gains.

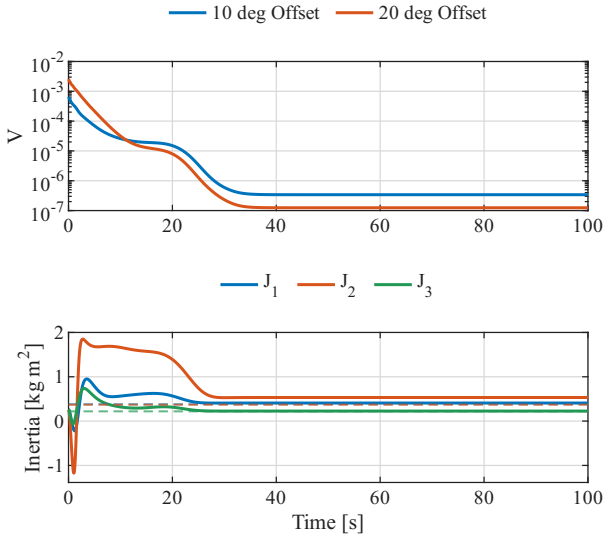


Fig. 6. Lyapunov function for the initial-error cases and inertia estimates for the 10° case. Dashed lines mark the true values.

E. Reference Motion and Inertia Estimation

Fig. 7 compares the oblique slew, principal-axis slew, and periodic motion at the nominal gains in Table III. The two slews have the same desired rate magnitude and hold time. The desired rate in the periodic case varies throughout the run.

At the nominal gains, the oblique slew brings all three estimates close to the true values, with a final error norm of 1.22×10^{-3} kg m^2. In the principal-axis run, the estimate of J_1 reaches 0.3799 kg m^2, and the estimates of J_2 and J_3 stay at their initial values to the precision shown in Table V. The periodic case brings all three estimates to the true values, with a final inertia error norm below 10^{-10} kg m^2.

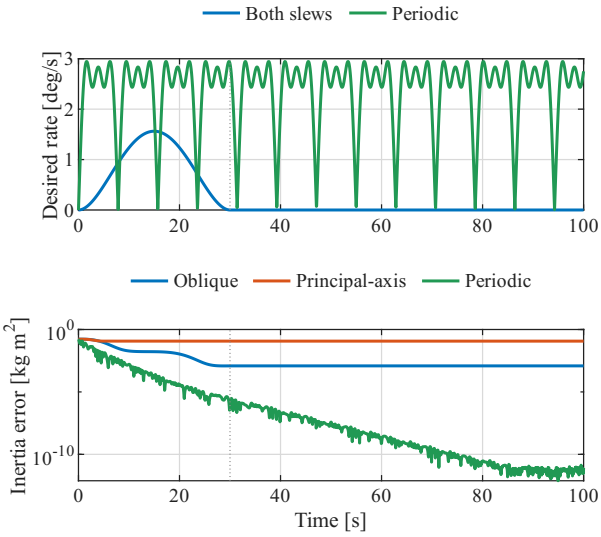


Fig. 7. Reference motion and inertia estimation at the nominal gains.

TABLE V
TRUE AND ESTIMATED MOMENTS OF INERTIA. ESTIMATES ARE EVALUATED AT THE END OF THE RUN.

Case	Estimate [kg m^2]		
	Axis 1	Axis 2	Axis 3
True inertia	0.380000	0.370000	0.220000
Oblique slew (nominal)	0.378935	0.369517	0.220340
Principal-axis slew	0.379919	0.260000	0.260000
Periodic motion	0.380000	0.370000	0.220000
10° offset	0.407433	0.532562	0.226468

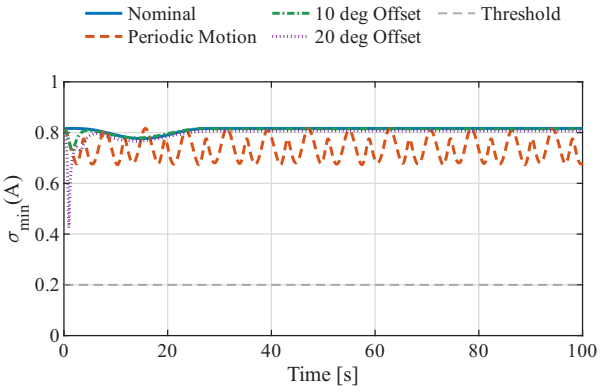


Fig. 8. CMG diagnostics for the nominal, periodic, and initial-error runs. The dashed line marks the singularity threshold.

F. CMG Feasibility and Torque Realization

The torque residual is defined as $\Delta\tau = \tau_{\text{cmg}} - \tau_{\text{cmd}}$ and is reported as $\|\Delta\tau\|$. Fig. 8 compares the smallest singular value of the steering Jacobian, $\sigma_{\min}(A)$, for the nominal, periodic, and initial-error runs. The torque residual and peak actuator demands for all runs are listed in Table VI.

The nominal run has a minimum singular value of 0.7763. The smallest value across all runs is 0.4245 in the 20° case, which remains above the 0.20 threshold in Table I. The largest torque residual across all runs is 1.74×10^{-16} N m.

TABLE VI
PEAK INPUT AND ACTUATOR DIAGNOSTICS OVER THE SIMULATED INTERVAL

Case	Peak torque norm [mN m]	Peak gimbal rate [deg/s]	Min. $\sigma_{\min}(A)$	Peak torque residual [N m]
Nominal	0.970	1.187	0.7763	1.01×10^{-18}
Lower K	0.997	1.217	0.7764	1.02×10^{-18}
Lower Γ	0.950	1.168	0.7754	1.32×10^{-18}
Lower α	0.972	1.190	0.7762	1.12×10^{-18}
10° offset	16.12	21.84	0.7277	1.30×10^{-17}
20° offset	61.99	124.6	0.4245	1.74×10^{-16}
Principal-axis slew	1.113	1.576	0.7894	6.54×10^{-19}
Periodic motion	16.71	24.05	0.6733	1.12×10^{-16}

VII. DISCUSSION

A. Effects of the Controller and Adaptation Gains

All four oblique-slew cases settle below 0.001° within 12 s of the end of the slew. A smaller K applies less feedback for a given auxiliary error r. The tracking peaks increase, while the peak commanded torque remains close to the nominal value (Table VI).

Reducing Γ decreases the magnitude of the inertia update for a given regressor and auxiliary error in (39). The estimates in this case remain far from the true values at 100 s, even though the tracking errors settle during the hold. Reducing α decreases the weight on the attitude error in r and also changes the regressor Y. The lower-α case has higher peak tracking errors than the nominal case, with nearly the same peak commanded torque.

The large nominal value of Γ is needed because the regressor is small for this maneuver. The slew lasts 30 s with a peak desired angular acceleration of about 0.0028 rad/s^2, so the entries of Y and the update rate in (39) are small unless Γ is large. With $\Gamma = 350 I_3$, each estimate changes by less than 0.005 kg m^2 over the run.

B. Initial Tracking Error and Parameter Estimates

The initial-error runs show the difference between tracking and parameter convergence. Both runs converge to the reference, as predicted by Theorem 1, but their estimates finish far from the true values. With $\Gamma = 40{,}000 I_3$, a large initial error produces a large regressor and auxiliary error at the start of the run, so the estimates change quickly before the tracking error decreases. The bound in (65) also grows with the adaptation gain, since $\lambda_{\min}(\Gamma^{-1}) = 1/40{,}000$. For the 10° case, $V_0 = 6.48 \times 10^{-4}$ and the bound on $\|\tilde{\theta}\|$ is about 7.2 kg m^2, so the analysis allows large estimate transients.

The update law has no constraint that keeps the estimates positive, and all three estimates become negative for a short time in the 10° case. A projection on the update law, as used in [8], would keep the estimates within known bounds. The tracking result does not depend on the estimates reaching the true values, so the tracking errors still converge.

C. Effects of the Reference Motion

During the principal-axis slew, the angular velocity and acceleration are concentrated about the first body axis. The terms in the second and third columns of the regressor in (35) therefore remain nearly zero, so the update acts only on the first inertia estimate. This is a limitation of the motion, and the nominal Γ does not recover J_2 and J_3 either. The oblique slew moves all three axes, and at the nominal gains it brings all three estimates close to the true values (Table V). Its excitation is small because the slew is slow and lasts only 30 s. After 30 s, the desired rate and acceleration are zero, and the estimates change very little during the hold.

The periodic reference uses a different sinusoidal frequency on each axis and keeps all three axes in motion for the full 100 s. Its peak desired acceleration is 0.042 rad/s^2 on the third axis, about 15 times the peak acceleration of the slews, and this excitation continues for the whole run. The periodic case therefore brings all three estimates to the true values, which is consistent with the periodic-motion example in Ahmed et al. [3]. A convergence guarantee for the inertia estimates would still require a separate analysis of excitation.

D. CMG Assumptions and Scope of the Results

The periodic motion requires greater actuator effort than the nominal slew. Its peak commanded torque is 16.71 mN m and its peak gimbal rate is $24.05°$/s, compared with 0.970 mN m and $1.187°$/s for the nominal slew (Table VI). The initial-error cases require much more effort at the start of the run. The $20°$ case reaches 61.99 mN m and $124.6°$/s, and $\sigma_{\min}(A)$ drops to 0.4245. At a singular configuration, the Jacobian loses rank, which limits the torque components that can be commanded through the gimbal rates [12, Sec. 5.2]. The singularity margin therefore depends on the size of the initial error and the gains, not only on the reference motion. The torque residuals are consistent with numerical roundoff in the pseudoinverse calculation.

The tracking errors observed in all runs are consistent with the asymptotic tracking result in Theorem 1. The simulations use constant flywheel speeds and ideal gimbal-rate tracking without actuator dynamics or gimbal-rate limits. The gimbal rates in the $20°$ case are much higher than in the other runs, and a gimbal with a rate limit may not deliver the commanded torque in that case. The singular value and torque residual checks support the assumptions for the tested trajectories over the simulated interval.

VIII. Conclusion

This project used a four-CMG array to study adaptive tracking with three unknown principal moments of inertia. The Lyapunov analysis established asymptotic attitude and angular velocity tracking and bounded inertia estimation error under exact torque realization and a uniform margin from CMG singularities. In simulation, the nominal oblique slew had the smallest peak tracking errors among the gain cases. Runs starting $10°$ and $20°$ from the desired attitude converged to the reference, while their inertia estimates finished away from the true values. At the nominal gains, the periodic motion brought all three estimates to the true values, while the principal-axis slew updated only the first moment. Across all runs, the lowest observed Jacobian singular value was 0.4245, above the 0.20 threshold, and the torque residuals were at numerical roundoff scale.

Limitations. The analysis assumes a rigid spacecraft with constant diagonal inertia. The ideal CMG model assumes exact gimbal-rate tracking and inactive rate and travel limits. The singularity and torque residual checks cover the tested 100 s trajectories. The small final inertia error in the periodic run is a finite-time simulation result. Parameter convergence requires a separate excitation analysis. The update law has no projection, so the estimates can become nonphysical during large transients.

Future work. A more detailed actuator model could include gimbal dynamics and enforce rate and travel limits. Maneuvers closer to singularity could be used to compare steering laws in terms of the torque residual and the minimum Jacobian singular value. The torque residual term in (95) gives a starting point for analyzing how imperfect torque realization affects tracking. A separate study could check whether the periodic reference provides sufficient excitation for convergence of all three inertia estimates. A projection on the update law [8] could keep the inertia estimates positive during large initial errors.

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